

PAMUDI MENDIS

14, Saddasena Rd, Ambalangoda | 070 272 2872 | pamudimendis@gmail.com | 

SUMMARY

Final-year Mechanical Engineering undergraduate at the University of Moratuwa with a strong foundation in engineering design, manufacturing, simulation, and naval and marine engineering. Experienced in academic, research, and industrial projects, with exposure to undergraduate teaching and laboratory instruction. Strong analytical, communication, and problem-solving skills, with a keen interest in research, innovation, and the practical application of engineering principles.

EDUCATION

University of Moratuwa	Mar 2022 - Present
B.Sc (Hons), Mechanical Engineering (CGPA - 3.46/4.00)	
IJSE (Institute of Software Engineering)	2020-2021
Comprehensive Master Java Developer	
Southlands College, Galle	2012 - 2020
- G.C.E. A/L 2020 : 3As - Physical Science Stream z-score - 2.6142, Island Rank - 90 - G.C.E. O/L 2017 : 9As (Gold Medal for the Best Academic Achiever - O/L)	

WORK EXPERIENCE

Visiting Instructor	Feb - May 2026
Department of Mechanical Engineering, University of Moratuwa	
- Conducted <i>Refrigeration and Air-conditioning</i> practical sessions to semester 2 students for the <i>Fundamentals of Engineering Thermodynamics</i> module. - Assisted in delivering <i>Mechanics</i> lectures for semester 1 students	
Trainee Mechanical Engineer	Dec 2024 - June 2025
Hyundai Assembly Plant - Abans Auto (Pvt) Ltd	
- Coordinated project planning and execution for assembly plant layout restructuring and warehouse mezzanine design - Conducted re-engineering analysis of the vehicle shower test system - Design of a wheel lifting machine for chassis assembly - Design of a detachable wheel mounting assembly to reduce forklift usage	

PROJECTS

Design and Development of a Metal Additive Manufacturing System for Complex Component Fabrication	July 2025 - Present
Designing and developing a metal 3D printer using WAAM technology with regulated control of key process parameters, including system modelling and simulation to predict behaviour, and experimental validation through fabrication of complex metal components.	
Production line optimization at Abans Electricals PLC	March - May 2026
Optimized the washing machine production line at Abans Electricals PLC by identifying inefficiencies and proposing improvements in material handling, batch sizing, and workflow.	

Concept Design of an Ambulance Craft for Operations in the Northern Islands, Sri Lanka

Sept - Dec 2025

Designed a concept ambulance craft, including hull modelling, stability analysis, general arrangement, and propulsion selection, while ensuring compliance with marine standards.

Development of a Below-Knee Prosthetic Leg Testing Machine

Sept - Dec 2025

Designed a low-cost, hand-operated mechanism to simulate gait cycles for static and fatigue testing of below-knee prosthetic limbs.

Social/community Project on the Sri Lankan Mold Manufacturing Industry

June - July 2025

Analyzed challenges in Sri Lanka's mold manufacturing industry through a case study on Hasala Engineering and proposed policy-level and technical solutions to improve global competitiveness.

Design of a Gearbox for a Domestic Coffee Grinder

July - Oct 2024

Designed a gearbox to enable multiple coffee grind sizes, optimizing the grinding process. Developed detailed 3D models and assemblies of the gearbox considering functionality and manufacturability.

Optimization of the Production Line of Wewalwala Mills (Pvt) Ltd

July - Oct 2024

Analyzed existing production line conditions and identified inefficiencies. Proposed simulation-based improvements to enhance productivity and overall production line efficiency.

Reverse Engineering of a Cloth Iron

Mar - July 2024

Identified materials and reverse-engineered all components of a domestic cloth iron. Developed 3D models using SolidWorks and performed FEA to recommend design improvements.

Manufacturing of a Laptop Table

Mar - July 2023

Fabricated a laptop table using welding, sheet metal fabrication, and wood-working processes. Gained hands-on experience in basic manufacturing techniques.

Smart Waste Management System

Mar - July 2023

Developed a smart waste management system using image classification on a Raspberry Pi for waste sorting. Integrated Arduino-based compression and sealing mechanisms, real-time monitoring, and waste generation pattern analysis.

LICENSES & CERTIFICATIONS

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| • <u>Sustainability for Design, Construction, and Manufacturing</u> | Issued Jul 2025 |
| • <u>Additive Manufacturing: Metal 3D Printing</u> | Issued Jul 2025 |
| • <u>Simulink Onramp</u> | Issued Jul 2024 |
| • <u>MATLAB Essential Training</u> | Issued Jul 2024 |
| • <u>MATLAB Onramp</u> | Issued June 2024 |

VOLUNTEERING EXPERIENCE

University of Moratuwa Mechanical Engineering Society - UMMES

Aug 2025 - Present

Secretary

- Contributed to UMMES initiatives through project coordination, people management, technical content development, social media management and event organization to enhance student engagement and professional development.

ECHELON'25

June - July 2025

Editorial Committee Lead

AIIESEC in University of Moratuwa

Oct 2022 - Feb 2024

- Organizing Committee Vice President - Marketing | Global Classroom 1.4 & Rooted 2.2
- Organizing Committee Member - Marketing | LEAD CS 10.0
- Organizing Committee Member - Partnership Development & Public Relations | HOPE 2023
- Organizing Committee Member - Events & Delegates | Global Village 2023
- Data Management Specialist - Finance & Legal Team

PUBLICATIONS

Design and Development of a Metal Additive Manufacturing System Using Metal Inert Gas (MIG) Technology (Extended Abstract)

- Presented at the Engineering Research Unit Symposium 2025. Published in the conference proceedings.
- Available at: [University of Moratuwa Digital Repository](#).

Towards Sustainable Metal Additive Manufacturing Using a Low-Cost MAG-Based WAAM System (Full Paper)

- Accepted for presentation at the 22nd Global Conference on Sustainable Manufacturing, September 2026.
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SKILLS

Programming Languages: Python, Java, C++, MATLAB

Software & Tools : SolidWorks, Solid Edge, ANSYS, AutoCAD, Siemens NX, MATLAB, Simulink

Languages: Sinhala, English

Soft Skills: Communication, People Management, Critical Thinking, Time Management, Fast Learner, Team Work

REFERENCES

Dr Thamasha Samarasinghe

Senior Lecturer

Department of Mechanical Engineering

University of Moratuwa.

thamashas@uom.lk

+94 77 745 1164

Dr. Eranga de Silva

Senior Lecturer

Department of Mechanical Engineering

University of Moratuwa.

ahteranga@uom.lk

+94 71 860 3738